Experiment 02

Lab Tasks

1. Create a comparison chart (hand-drawn or digital) of frontend vs backend: 5 differences, 3 tools each, 1 example job ad for each from rozee.pk.
2. Write valid JSON describing yourself: name, university, semester, 3 skills (array), and social links (nested object). Validate it on jsonlint.com and attach a screenshot.
3. Use the browser console fetch() to call https://api.github.com/users/github and list any 5 fields returned with their values.
4. Pick any Pakistani app (Careem, Foodpanda, Easypaisa) and write a one-page story of what happens across frontend → API → backend → database when a user performs one action.
5. List the full set of technologies in the MERN stack and one sentence on what each does.

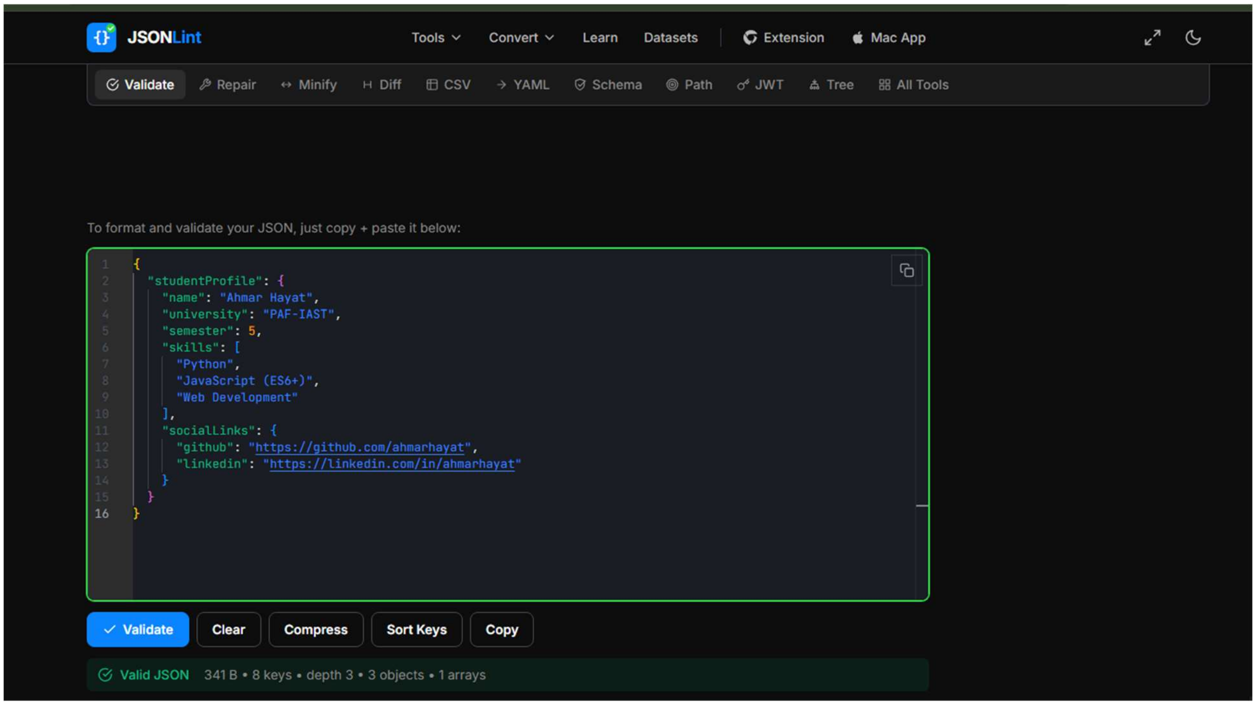
1. Create a comparison chart (hand-drawn or digital) of frontend vs backend: 5 differences, 3 tools each, 1 example job ad for each from rozee.pk.

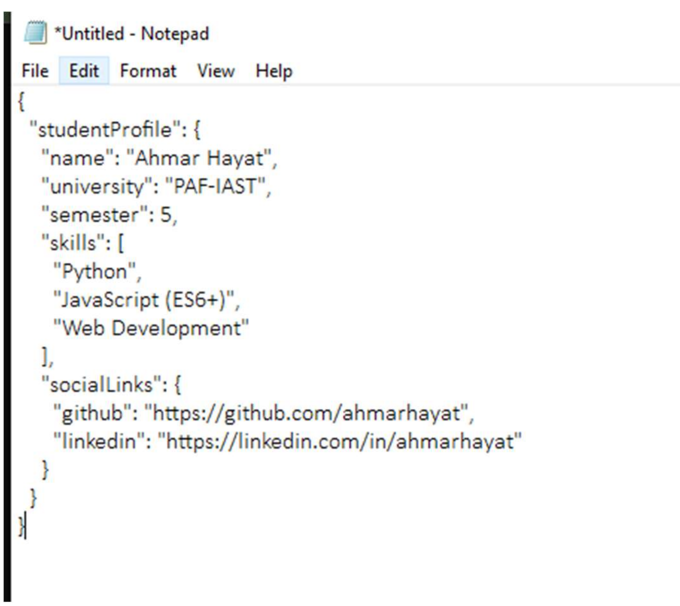
Criteria	Frontend Development	Backend Development
Definition	Everything the user visualizes and directly interacts with in the web browser (Client-Side).	The background server infrastructure, business logic, and database layer (Server-Side).
Core Focus	User Experience (UX), layout structure, client-side state, responsiveness, animation.	Authentication, authorization, database CRUD, security algorithms, REST APIs.
Execution	Runs inside the user's web browser (Chrome, Firefox, Edge).	Runs on web servers or cloud runtime environments (Node.js, AWS, Vercel).
Data Handling	Consumes JSON data via APIs and transforms it into formatted HTML/CSS.	Fetches data from database tables, transforms it into JSON, and serves client requests.
Security Risks	Cross-Site Scripting (XSS), exposed API keys, UI bypass.	SQL Injection, broken access control, CORS misconfigurations, DDoS attacks.

Essential Frontend Tools: React.js, Tailwind CSS, VS Code.

Essential Backend Tools: Node.js, Express.js, MongoDB.

2. Write valid JSON describing yourself: name, university, semester, 3 skills (array), and social links (nested object). Validate it on jsonlint.com and attach a screenshot.

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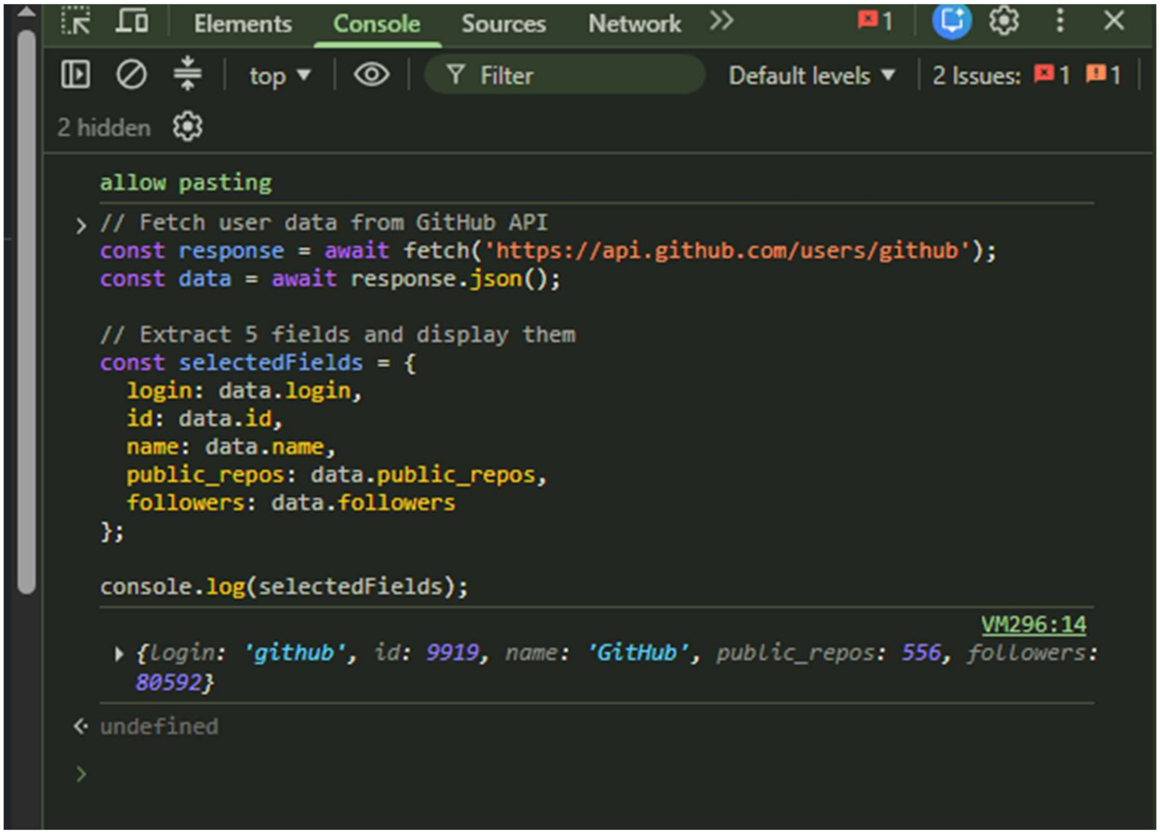
```
{
  "studentProfile": {
    "name": "Ahmar Hayat",
    "university": "PAF-IAST",
    "semester": 5,
    "skills": [
      "Python",
      "JavaScript (ES6+)",
      "Web Development"
    ],
    "socialLinks": {
      "github": "https://github.com/ahmarhayat",
      "linkedin": "https://linkedin.com/in/ahmarhayat"
    }
  }
}
```

3. Use the browser console fetch() to call https://api.github.com/users/github and list any 5 fields returned with their values.

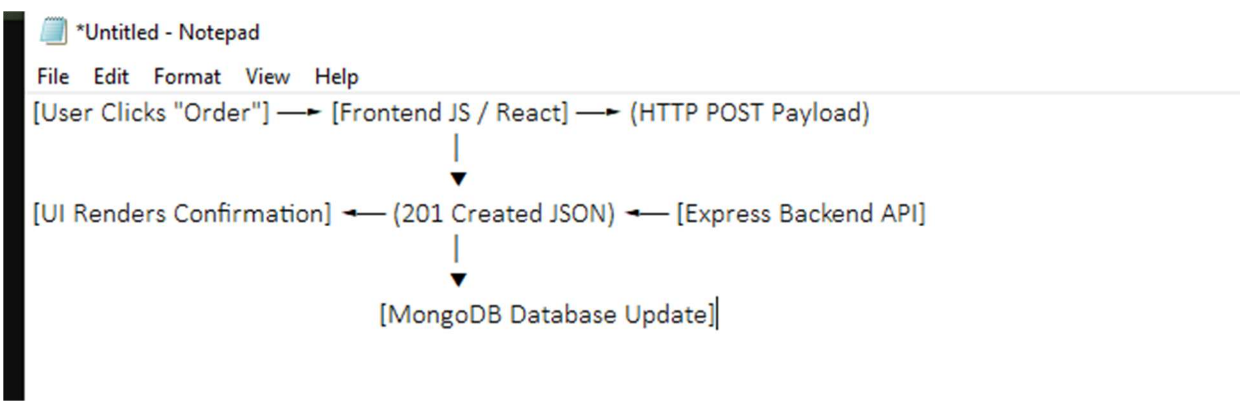


```
allow pasting
> // Fetch user data from GitHub API
const response = await fetch('https://api.github.com/users/github');
const data = await response.json();

// Extract 5 fields and display them
const selectedFields = {
  login: data.login,
  id: data.id,
  name: data.name,
  public_repos: data.public_repos,
  followers: data.followers
};

console.log(selectedFields);
                                                              VM296:14
  ▸ {login: 'github', id: 9919, name: 'GitHub', public_repos: 556, followers: 80592}
< undefined
```

4. Pick any Pakistani app (Careem, Foodpanda, Easypaisa) and write a one-page story of what happens across frontend → API → backend → database when a user performs one action.



```
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[User Clicks "Order"] ──► [Frontend JS / React] ──► (HTTP POST Payload)
                                  |
                                  ▼
[UI Renders Confirmation] ◄── (201 Created JSON) ◄── [Express Backend API]
                                  |
                                  ▼
                         [MongoDB Database Update]
```

1. **User Action (Frontend):** A user in Haripur selects a burger on Foodpanda and taps **"Place Order"**. The client-side React component gathers item IDs, quantities, address coordinates, and user authentication tokens.
2. **API Request Dispatch:** The browser sends an asynchronous HTTPS POST request with a JSON payload to https://api.foodpanda.pk/v1/orders.
3. **Backend Processing:** The Node.js / Express backend intercepts the incoming HTTP request. It validates the user's JWT token, executes business logic to confirm item prices, and calculates delivery fees.
4. **Database Mutation:** The backend opens a transaction with MongoDB, inserting a new record into the orders collection and updating restaurant inventory counts.
5. **JSON Response & UI Update:** MongoDB confirms the database write. The server responds with status code 201 Created and a JSON body containing { "orderId": "FP-9821", "estimatedTime": "25 mins" }. The frontend receives the payload and renders a live delivery tracking animation screen.

5. List the full set of technologies in the MERN stack and one sentence on what each does.

MERN Stack Component Overview

- **MongoDB:** A NoSQL document-based database that stores data in flexible, JSON-like document collections.
- **Express.js:** A lightweight backend application framework built for Node.js to streamline HTTP route handling and middleware creation.
- **React.js:** A declarative, component-driven JavaScript library for constructing reactive web interfaces.
- **Node.js:** An open-source asynchronous JavaScript runtime environment that executes JS code server-side outside the browser.
